In [1]:
```
# Q3 I. a)
```

In [14]:
```
import numpy as np
from scipy.optimize import fsolve

def f(x):
    return x**2 - np.exp(-x) - 2 * np.sin(x)

initial_guess = 1.0
root = fsolve(f, initial_guess)

print("Root of f(x):", root[0]) #The fsolve() function always returns a NumPy array
```

```
Root of f(x): 1.4895886408077366
```

In [3]:
```
# Q3 I. b)
```

In [15]:
```
import numpy as np
from scipy.optimize import fsolve

def f(t):
    return 10 * t - 4.9 * t**2

initial_guess = 2.0
root = fsolve(f, initial_guess)

print("Time when projectile returns to initial height:", root[0], "seconds")
```

```
Time when projectile returns to initial height: 2.040816326530673 seconds
```

In [8]:
```
# Q3 I. c)
```

In [16]:
```
import numpy as np
from scipy.optimize import fsolve

def f(x):
    return x**2 - 3

initial_guess = 1.5
root = fsolve(f, initial_guess)

print("Approximate sqrt(3):", root[0])
```

```
Approximate sqrt(3): 1.7320508075688725
```

In [11]:
```
# Q3 I. d)
```

In [17]:
```
import numpy as np
from scipy.optimize import fsolve

def f(L):
    return 2 * np.pi * np.sqrt(L / 9.8) - 10

initial_guess = 50
```

```
root = fsolve(f, initial_guess)

print("Length L for T = 10 seconds:", root[0], "meters")
```

```
Length L for T = 10 seconds: 24.823689992373527 meters
```

In [ ]:
```
# Q3 I. e)
```

In [18]:
```
import numpy as np
from scipy.optimize import fsolve

def f(t):
    return 5 * (1 - np.exp(-t / 10)) - 2.5

initial_guess = 10
time = fsolve(f, initial_guess)

print("Time at which Vc(t) = 2.5 V:", time[0], "seconds")
```

```
Time at which Vc(t) = 2.5 V: 6.931471805599454 seconds
```

In [ ]: